

Sephora Product & Customer Review Analytics

End-to-End Data Analytics Portfolio Project

1. Project Overview

Build an end-to-end analytics solution using the Sephora Products and Skincare Reviews dataset. The project combines Python for data cleaning and sentiment analysis, SQL for business analysis, and Power BI for executive dashboards that support merchandising and product decisions.

2. Business Problem

Thousands of products and customer reviews make it difficult to identify underperforming brands, understand customer dissatisfaction, evaluate pricing strategy, and prioritize products requiring business attention.

3. Objectives

- Identify top and bottom performing products and brands.
- Analyze customer sentiment.
- Evaluate price versus rating.
- Detect high-risk products.
- Deliver executive dashboards and actionable recommendations.

4. Scope

In Scope:

- Data cleaning (Python)
- SQL business analysis
- Sentiment analysis
- Power BI dashboard
- Business recommendations

Out of Scope:

- Sales forecasting
- Inventory optimization
- Marketing analytics

5. Stakeholders

Executive Leadership, Merchandising Team, Product Managers, Brand Managers, Customer Experience Team, Data Analyst.

6. Business Questions

Which categories and brands perform best? Which products have low ratings but high review volume? Does price influence ratings? What drives negative sentiment? Which products require immediate action?

7. Deliverables

- Clean dataset
- SQL scripts
- Python notebooks
- Sentiment analysis
- Power BI dashboard
- Documentation
- GitHub repository

8. Technology Stack

SQL, Python, Pandas, NLP, Power BI, DAX, Power Query, Git, GitHub.

9. Success Criteria

Provide a decision-ready dashboard, identify business risks, generate actionable insights, and support data-driven product and merchandising decisions.